

2 The basic framework

In this section we state and explain the essential definitions of probability theory: sample space, event, probability measure, conditional probability, independence, random variable and its distribution. Most further concepts in probability theory, even at an advanced level, can be expressed in terms of these basic notions. While systematic investigation of probability theory may be said to begin with the work of Pascal and Fermat in the 17th century, the modern definitions that we adopt here were established fairly recently by Kolmogorov in 1933. It will take a while to understand and develop an appreciation of the mathematical structure.

2.1 Sample space

An *experiment* is a chance event or process that we want to model using probability theory. We assume that there is a (nonempty) *set* Ω containing all possible outcomes of the experiment. We call Ω the *sample space*. We usually use ω to denote a generic element of Ω . An element $\omega \in \Omega$ is called a *sample point*; it represents a specific outcome of the experiment.

To describe sample spaces accurately and compactly we need some notations from *set theory*. Here are some examples.

Example 2.1 (Examples of sample spaces).

- (i) Tossing a coin once. We may pick $\Omega = \{0, 1\}$ where 0 means tails and 1 means heads (say).
- (ii) Rolling a dice once. We may pick $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- (iii) Rolling a dice twice. We may pick

$$\Omega = \{(1, 1), \dots, (1, 6), \dots, (6, 1), \dots, (6, 6)\} = \{(i, j) : i, j = 1, 2, \dots, 6\}.$$

- (iv) Recall the concept of *product set*. Given sets A and B , the *product set* $A \times B$ is defined as a set of *ordered pairs*:

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

Similarly, we may consider the product of a finite or infinite sequence of sets A_i , $i = 1, 2, \dots$:

$$\prod_{i \geq 1} A_i = A_1 \times A_2 \times \dots = \{(a_1, a_2, \dots) : a_1 \in A_1, a_2 \in A_2, \dots\}.$$

If $A_1 = A_2 = \dots = A$, we denote the *n-fold product* by

$$A^n = \underbrace{A \times \dots \times A}_{n \text{ times}}.$$

For example, if we roll a dice n times, we may let

$$\Omega = \{1, 2, \dots, 6\}^n = \{\omega = (\omega_1, \omega_2, \dots, \omega_n) : \omega_i \in \{1, 2, \dots, 6\}\}.$$

Each sample point is a sequence with values in $\{1, 2, \dots, 6\}$ and has length n . This sample space has $|\Omega| = 6^n$ elements.

- (v) Tossing a coin infinitely many times. (This is theoretically useful even though in reality we cannot toss a coin infinitely many times.) We may let $\Omega = \{0, 1\}^\infty$ be the infinite product $\{0, 1\} \times \{0, 1\} \times \cdots$ which is a *sequence space*. By definition, each ω is a sequence of the form $(\omega_1, \omega_2, \dots)$ where $\omega_i \in \{0, 1\}$.
- (vi) Measuring the direction of a radar signal. We may consider the continuous interval $\Omega = [0, 2\pi)$, where each $\omega \in \Omega$ is interpreted as an angle (measured in radians).

2.2 Events

Given a sample space Ω , an *event* is a subset of Ω . Given an outcome $\omega \in \Omega$, we say that an event A *occurs* if $\omega \in A$. Otherwise A does not occur. Two special events are:

- The *impossible event* represented by the empty set $\emptyset$. It never occurs.
- The *certain event* represented by the whole space Ω . It always occurs.

Although we often describe an event informally in words, it must correspond to a well-defined subset of the sample space.

Example 2.2. Suppose we roll two dice as in Example 2.1(iii). Let A be the event that the sum of the two numbers is 9. By enumeration, we see that

$$A = \{(3, 6), (4, 5), (5, 4), (6, 3)\}.$$

Example 2.3. Suppose we obtain two measurements with values in $[0, 1]$. Let $\Omega = [0, 1]^2$ be the unit square. Let A be the event that the first measurement is at least 0.3 *and* the second measurement is greater than or equal to the first one. Then

$$A = \{\omega = (\omega_1, \omega_2) \in \Omega : \omega_1 \geq 0.3 \text{ and } \omega_2 \geq \omega_1\}.$$

We can create new events from given ones using *logical operations*. These operations correspond closely to *set operations* such as union and intersection. In the following we let $A, B, A_i, \dots$ be events in a given sample space Ω .

- The event that A does not occur is the *complement* A^c .
- The event that A or B (or both) occurs is the *union* $A \cup B$. Similarly, given a (finite or infinite) sequence $(A_i)_{i \geq 1}$ of events, the event that at least one of them occurs is the union $\bigcup_{i \geq 1} A_i$.
- The event that both A and B occur is the *intersection* $A \cap B$. Similarly, we may consider the intersection $\bigcap_{i \geq 1} A_i$ for a given sequence of events.

Other set operations can be expressed in terms of complement, union and intersection. For example:

- The event that A occurs but not B is the *relative complement* $A \setminus B = A \cap B^c$.

- The event that one and only one of A or B occurs is the *symmetric difference* $A \triangle B = (A \setminus B) \cup (B \setminus A)$.

We note the following useful properties of set operations:

- *Negation is involutive:* $(A^c)^c = A$.
- *De Morgan's laws:* $(\bigcup_i A_i)^c = \bigcap_i A_i^c$ and $(\bigcap_i A_i)^c = \bigcup_i A_i^c$.
- *Distributive properties:* $B \cap (\bigcup_i A_i) = \bigcup_i (B \cap A_i)$ and $B \cup (\bigcap_i A_i) = \bigcap_i (B \cup A_i)$.

These operations and properties may be visualized using *Venn diagrams*. Apart from creating new events from given ones, they are also useful for *decomposing a complicated events in terms of simpler ones*.

Events $A_1, A_2, \dots$ are said to be *disjoint*, or *mutually exclusive*, if $A_i \cap A_j = \emptyset$ whenever $i \neq j$. In view of the sum rule (see (2.5) below), it is often useful to decompose an event as a disjoint union of simpler events.

Example 2.4. Continuing with Example 2.1(iii), suppose we are interested in the sum of the two numbers. Then

$$\{\text{sum} \geq 10\} = \{\text{sum} = 10\} \cup \{\text{sum} = 11\} \cup \{\text{sum} = 12\}$$

and the union is disjoint. As an exercise, you should *visualize* these sets when the elements of $\Omega = \{1, \dots, 6\}^2$ are represented by a 6×6 matrix.

The mathematically inclined student may read Remark A.1 (in Section A.2) at this point.

2.3 Probability measure

A *probability measure* defines the probabilities of various events. Before stating the general definition we give some motivating examples.

Example 2.5. In the example illustrated in Table 1.1, we began by defining the probabilities of individual outcomes. It was implicitly understood that the probability of an event A is equal to the sum of the probabilities of the sample points in A . For example, we have

$$\begin{aligned} \mathbb{P}(X = 2) &= \mathbb{P}\{++-, +-+, -++\} \\ &= \mathbb{P}\{++-\} + \mathbb{P}\{+-+\} + \mathbb{P}\{-++\} \\ &= 3 \cdot \frac{1}{8} = \frac{3}{8}. \end{aligned} \tag{2.1}$$

This construction ensures that the *sum rule* mentioned in Section 1.2 holds.

The construction in Example 2.5 can be vastly generalized. Say that a set A is *discrete* (or *countable*) if A is either finite or countably infinite. By definition, A is *countably infinite* if we may list all distinct elements of A in terms of a sequence: $A = \{a_1, a_2, \dots\}$. In other words, the elements of A are in 1-to-1 correspondence with the elements of $\mathbb{N} = \{1, 2, \dots\}$, the set of natural numbers.¹ For example, the set $\{2, 4, \dots\}$ of even numbers is countable.

¹The alternative convention $\mathbb{N} = \{0, 1, 2, \dots\}$ is also in common use.

Example 2.6 (Defining probabilities on a discrete sample space). Let $\Omega = \{\omega_1, \omega_2, \dots\}$ be a discrete sample space. Suppose for each i we are given a number $p_i \geq 0$, such that $\sum_{i=1}^{\infty} p_i = 1$ (depending on the situation we may also let the index i start from 0). First, define the probability of the singleton $\{\omega_i\}$ by $\mathbb{P}\{\omega_i\} = p_i$.² Next, for any event $A \subset \Omega$, we define

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}\{\omega\}. \quad (2.2)$$

Again the sum rule holds by construction: if $A_1, A_2, \dots$ are disjoint, then

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{\omega \in \bigcup_{i \geq 1} A_i} \mathbb{P}\{\omega\} = \sum_{i \geq 1} \sum_{\omega \in A_i} \mathbb{P}\{\omega\} = \sum_{i \geq 1} \mathbb{P}(A_i).^3$$

To give a concrete example, suppose $\Omega = \mathbb{N} = \{1, 2, \dots\}$. For each $i \in \mathbb{N}$, let $p_i = 2^{-i}$. Note that $\sum_{i=1}^{\infty} 2^{-i} = 1$, so we may use (2.2) to define the probability $\mathbb{P}(A)$ for any A . (This is a special case of the *geometric distribution* which will be introduced later.) Suppose $A = \{2, 4, \dots\}$ is the set of even numbers. Using the *geometric sum formula*

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}, \quad (2.3)$$

which holds for $|r| < 1$, we have

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} 2^{-2i} = \sum_{i=1}^{\infty} 4^{-i} = \frac{1}{4} \sum_{i=0}^{\infty} 4^{-i} = \frac{1}{4} \frac{1}{1-1/4} = \frac{1}{3}. \quad (2.4)$$

The following definition takes the sum rule as an *axiom*. It does not tell us how $\mathbb{P}$ is constructed; rather, it tells us how $\mathbb{P}$ must behave. Here, Kolmogorov's main contribution is to identify exactly *which* properties to put in the definitions so that a consistent and far-reaching theory can be built.

Definition 2.7 (Probability measure and probability space). *A probability measure (sometimes also called a probability distribution) $\mathbb{P}$ on Ω associates to each event A a non-negative real number $\mathbb{P}(A)$, to be called the probability of A , such that the following properties hold:*

- (i) (Normalization) $\mathbb{P}(\Omega) = 1$.
- (ii) (Sum rule) If $A_1, A_2, \dots$ is a (finite or infinite) sequence of disjoint events, then

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i). \quad (2.5)$$

²For notational simplicity, when $A = \{\dots\}$ we frequently write $\mathbb{P}(A) = \mathbb{P}\{\dots\}$ instead of $\mathbb{P}(\{\dots\})$.

³This depends on the fact that in a series with non-negative terms the terms can be rearranged without affecting the sum.

A probability space is a sample space equipped with a probability measure.⁴

In words, a probability measure is a set function $\mathbb{P}(\cdot)$ which takes as input an event A and returns the probability $\mathbb{P}(A)$ of the event. Intuitively, a probability measure $\mathbb{P}$ distributes a total mass of 1 across the sample space Ω . The probability $\mathbb{P}(A)$ of an event A is the amount of mass assigned to A .

Example 2.8. Any $\mathbb{P}$ constructed as in Example 2.5 is a probability measure.

We illustrate how these axioms can be used to deduce a number of useful properties. While they may be intuitively obvious, we emphasize that they are logical consequences of the axioms. Also, they are universal properties which hold for *all* probability measures (so they are always available). In the following we let a probability space $(\Omega, \mathbb{P})$ be given.

Proposition 2.9 (Elementary properties). *For any probability space $(\Omega, \mathbb{P})$ the following properties hold:*

- (i) *(Probability of complement) For any event A we have $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. In particular, we have $\mathbb{P}(\emptyset) = 0$.*
- (ii) *(Monotonicity) If A and B are events such that $A \subset B$ (i.e., A implies B), then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*
- (iii) *(Probability of union) For any events A and B we have $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. In particular, we have $\mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.*

Proof. (i) Consider $A_1 = A$ and $A_2 = A^c$ which form a disjoint sequence of events. Since $A \cup A^c = \Omega$, we have, by the sum rule,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

Hence $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. When $A = \Omega$, $A^c = \emptyset$. So $\mathbb{P}(\emptyset) = 1 - 1 = 0$.

(ii) Suppose $A \subset B$. Note that $B = A \cup (B \setminus A)$ expresses B as a disjoint union. By the sum rule, we have

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A).$$

(iii) Consider the disjoint unions

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A),$$

$$A = (A \setminus B) \cup (A \cap B) \quad \text{and} \quad B = (A \cap B) \cup (B \setminus A).$$

Applying the sum rule again, we have

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) \\ &= (\mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B)) + (\mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A)) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B). \end{aligned}$$

The general form of this identity, for a finite union $\bigcup_{i=1}^n A_i$, is called the *inclusion-exclusion principle*. $\square$

⁴Strictly speaking, a probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a sample space, $\mathcal{F}$ is a σ -algebra of events on Ω (see Remark A.1), and $\mathbb{P}$ is a probability measure defined for events in $\mathcal{F}$. For simplicity, we omit $\mathcal{F}$ and denote a probability space by $(\Omega, \mathbb{P})$.

Remark 2.10 (Union bound). The inequality $\mathbb{P}(\bigcup_i A_i) \leq \sum_i \mathbb{P}(A_i)$ holds for any sequence $(A_i)_{i \geq 1}$ of events. This *union bound*, also known as *Boole's inequality*, is often used to give an upper estimate of the probability of an event. For example, suppose a student answers two questions and the probability of giving an incorrect answer is 0.1 for each question. Then the probability that some answer is incorrect is at most 0.2, even if the two questions (and answers) are related in some way.

In probability theory it is often helpful to distinguish between *discrete* and *continuous* probability distributions. The discrete case has been discussed in Example 2.6. Here we give a basic example of a continuous probability distribution. The precise meaning of the word “continuous” will be discussed later. We also stress that there are probability distributions which are neither discrete nor continuous.

Example 2.11 (Uniform distribution on an interval). Consider the experiment of choosing a number between 0 and 1 “uniformly at random”. Let the sample space be $\Omega = [0, 1]$. Consider an interval $[a, b]$ where $0 \leq a < b \leq 1$. We expect that the probability of observing ω in the interval should be proportional to the length $b - a$ of the interval. Since the total length of Ω is 1, we should define

$$\mathbb{P}([a, b]) = b - a, \quad 0 \leq a < b \leq 1. \quad (2.6)$$

For example, we have $\mathbb{P}([\frac{1}{4}, \frac{3}{4}]) = \frac{3}{4} - \frac{1}{4} = \frac{2}{4}$. It can be shown that (2.6) defines consistently and uniquely a probability measure, still denoted by $\mathbb{P}$, on $[0, 1]$. We call this $\mathbb{P}$ the *uniform distribution* on $[0, 1]$. Intuitively, the probability $\mathbb{P}(A)$ of a set $A \subset [0, 1]$ is simply the “length” of A .

The uniform distribution has the property that for any $x \in [0, 1]$ we have

$$\mathbb{P}(\{x\}) = 0, \quad (2.7)$$

even though the experiment must result in a number on $[0, 1]$. This follows from (2.6) by letting $a \leq x \leq b$ and sending $b - a \rightarrow 0$: $\mathbb{P}(\{x\}) \leq \mathbb{P}([a, b]) = b - a \rightarrow 0$. In short, a point has zero length. The subtlety is that the sample space $[0, 1]$ is not discrete and is in fact *uncountably infinite* (this means that the elements of $[0, 1]$ cannot be enumerated through a sequence). This implies that continuous distributions cannot be described by a probability mass function; rather, one must use a probability *density* function which will be introduced in Definition 4.12.

2.4 Conditional probability

While the sum rule deals with the probability of a (disjoint) union, the *product rule* deals with the probability of an intersection.

Definition 2.12 (Conditional probability). Let A, B be events. If $\mathbb{P}(B) > 0$, we define the conditional probability of A given B by

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (2.8)$$

If $\mathbb{P}(B) = 0$ we leave $\mathbb{P}(A|B)$ undefined.

The operational meaning of (2.8) is the *product rule* which takes the form

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A|B). \quad (2.9)$$

When $\mathbb{P}(B) = 0$, we have automatically $\mathbb{P}(A \cap B) = 0$ since $A \cap B \subset B$. Since the multiple of any finite number and zero is zero, by an abuse of notations we may still use (2.9) even when $\mathbb{P}(B) = 0$ and $\mathbb{P}(A|B)$ is undefined (this is because both sides of (2.9) are zero regardless how we define $\mathbb{P}(A|B)$). This convention will be handy in many computations. An example is the *law of total probability* to be discussed in Section 3.1.

Remark 2.13 (Frequentist interpretation of conditional probability). Suppose we throw two fair dice and their sum is 6. Let's consider the "frequentist conditional probability" $\Pr(\text{first die is 1}|\text{sum is 6})$.

Note that (1.1) cannot be applied directly in this case because the sum of the two dice is not always 6, so the denominator n is misleading. The idea is that the condition "the sum is 6" changes the "effective sample space" of the experiment.

Suppose we repeat the experiment n times. Out of these n trials, suppose in $n_1 \leq n$ of them the sum is 6. The remaining $n - n_1$ trials are irrelevant. We are led to consider the following relative frequency:

$$\frac{\# \text{ of times the first die is 1 (out of the trials where the sum is 6)}}{\# \text{ of times the sum is 6}}.$$

The conditional probability $\Pr(\text{first die is 1}|\text{sum is 6})$ is then the limiting value of this fraction. Dividing the nominator and denominator by n , we have

$$\begin{aligned} & \frac{(\# \text{ of times the first die is 1 (and the sum is 6)})/n}{(\# \text{ of times the sum is 6})/n} \\ & \rightarrow \frac{\Pr(\{\text{first die is 1}\} \cap \{\text{sum is 6}\})}{\Pr(\{\text{sum is 6}\})}, \end{aligned}$$

as n tends to infinity. This limit motivates the definition (2.8).

We complete the example by computing the numerical value of the conditional probability. We have

$$\{\text{sum is 6}\} = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}.$$

Since $|\Omega| = 36$ and $\mathbb{P}$ is uniform on Ω by assumption, we have $\mathbb{P}(\{\text{sum is 6}\}) = \frac{5}{36}$ and $\mathbb{P}(\{\text{first die is 1}\} \cap \{\text{sum is 6}\}) = \frac{1}{36}$. So

$$\mathbb{P}(\{\text{first die is 1}\}|\{\text{sum is 6}\}) = \frac{1/36}{5/36} = \frac{1}{5}.$$

Note that this is slightly larger than the *unconditional probability*

$$\mathbb{P}(\{\text{first die is 1}\}) = \frac{1}{6}.$$

□

Before giving the Bayesian interpretation, we note that the product rule continues to hold after conditioning. (This also follows from Proposition 2.16 below.)

Lemma 2.14 (Product rule). *For any events A , B and C with $\mathbb{P}(C) > 0$, we have*

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C \cap A). \quad (2.10)$$

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B|C) &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} \quad (\text{by (2.8)}) \\ &= \frac{\mathbb{P}(B|A \cap C)\mathbb{P}(A \cap C)}{\mathbb{P}(C)} \quad (\text{by the product rule}) \\ &= \mathbb{P}(B|A \cap C)\mathbb{P}(A|C). \quad (\text{by (2.8) again}) \end{aligned}$$

□

Remark 2.15 (Bayesian interpretation of the product rule). From the Bayesian perspective, we think of $\mathbb{P}(A|I)$ as the plausibility of the event A given the available information I .

Consider the plausibility of the event $A \cap B$ (given I). In order that A and B is true, it must hold that A is true. Thus, we first reason about the plausibility of A . This gives $\mathbb{P}(A|I)$. Next, we should reason about the plausibility of B given that A (and I) holds. This gives $\mathbb{P}(B|I \cap A)$. Now (2.10) states that the plausibility of A and B (given I) can be found from the previous two probabilities by forming a product:

$$\mathbb{P}(A \cap B|I) = \mathbb{P}(B|I \cap A)\mathbb{P}(A|I).$$

On the other hand, we may first reason about the plausibility of B given I , and then A given B and I . This gives the alternative product

$$\mathbb{P}(A|I \cap B)\mathbb{P}(B|I).$$

The definition of conditional probability, and the product rule, ensure that both approaches yield the same conditional probability:

$$\mathbb{P}(B|I \cap A)\mathbb{P}(A|I) = \mathbb{P}(A \cap B|I) = \mathbb{P}(A|I \cap B)\mathbb{P}(B|I).$$

Thus the product rule amounts to *consistency* of the rules for plausible reasoning. In fact, it can be shown (see the discussion in Section 1.1.2) that the sum and product rules are, in a suitable sense, the *unique* rules which allow consistent plausible reasoning.

The next proposition says that conditioning induces a new probability measure on the sample space.

Proposition 2.16 (Conditional probability as a probability measure). *Let B be an event with $\mathbb{P}(B) > 0$. Then the conditional probability $\mathbb{P}(A|B)$, as a function of A , defines a probability measure on Ω .*

Proof. We verify that $\mathbb{P}(\cdot|B)$ satisfies the properties in Definition 2.7.

(i) (Normalization) $\mathbb{P}(\Omega|B) = \frac{\mathbb{P}(\Omega \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1$.

(ii) (Sum rule) Let $(A_i)_{i \geq 1}$ be a sequence of disjoint events. Then so is $(A_i \cap B)_{i \geq 1}$. We have

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i \geq 1} A_i \middle| B\right) &= \frac{\mathbb{P}((\bigcup_{i \geq 1} A_i) \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\bigcup_{i \geq 1} (A_i \cap B))}{\mathbb{P}(B)} \\ &\stackrel{\text{sum rule}}{=} \sum_{i \geq 1} \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_{i \geq 1} \mathbb{P}(A_i|B). \end{aligned}$$

□

2.4.1 Independence and conditional independence

Conditioning generally changes the probability of an event. When the conditional probability $\mathbb{P}(A|B)$ is the same as the unconditional probability $\mathbb{P}(A)$, the two events involved are said to be *independent*. The concept of independence is of profound importance in probability theory. Here we only state the definition for two events and give some elementary examples. We will continue the study of independence in later sections.

Definition 2.17 (Independence (for two events)). *Two events A and B are said to be independent, with respect to $\mathbb{P}$, if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \quad (2.11)$$

We write $A \perp\!\!\!\perp B$ (or $A \perp\!\!\!\perp_{\mathbb{P}} B$ if we want to make $\mathbb{P}$ explicit).

Suppose A and B are independent. If $\mathbb{P}(B) > 0$, from (2.11) we have

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A). \quad (2.12)$$

Similarly, we have $\mathbb{P}(B|A) = \mathbb{P}(B)$ if $\mathbb{P}(A) > 0$. Conversely, if either $\mathbb{P}(A|B) = \mathbb{P}(A)$ or $\mathbb{P}(B|A) = \mathbb{P}(B)$, then (2.11) holds and so A and B are independent. Intuitively, independence means that the information from B does not affect our probabilistic inference of A and vice versa.

Example 2.18. Suppose A and B are disjoint events with strictly positive probabilities. Then A and B are dependent. To prove this from the definition, we simply note that

$$\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0 \quad \text{but} \quad \mathbb{P}(A)\mathbb{P}(B) > 0. \quad (2.13)$$

We explain the intuition behind (2.13). Since A and B are disjoint, if B occurs then A cannot occur. So $\mathbb{P}(A|B) = 0 \neq \mathbb{P}(A)$ which contradicts (2.12) (which would hold if A and B were independent). Similarly, $\mathbb{P}(B|A) = 0 \neq \mathbb{P}(B)$.

Example 2.19. Consider the situation depicted in Table 1.1. Let

$$\begin{aligned} A &= \{\text{second toss gives } +\} = \{+++, ++-, -++, -+-\}, \\ B &= \{\text{third toss gives } +\} = \{+++, +-+, -++, ---\}. \end{aligned}$$

We have $A \cap B = \{+++, -++\}$. Since $\mathbb{P}$ is uniform on Ω , counting gives

$$\mathbb{P}(A \cap B) = \frac{2}{8} = \frac{4}{8} \cdot \frac{4}{8} = \mathbb{P}(A)\mathbb{P}(B).$$

So A and B are independent.

As an exercise, you should verify that if C is the event that the second and third tosses are different, then C and A , as well as C and B , are independent. So A , B and C are *pairwise independent*. However, it is intuitively clear that they are not *jointly* independent (the precise definition will be given later): for example, if both A and B occur then C cannot occur.

The relation between two events may change according to the available information. Intuitively, *unconditional* independence as in Definition 2.17 refers to independence without any information, e.g. “before the experiment”. More generally, we have the following definition:

Definition 2.20 (Conditional independence). *Let C be an event with $\mathbb{P}(C) > 0$. Two events A and B are said to be conditionally independent given C if*

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C). \quad (2.14)$$

From Proposition 2.16, $\mathbb{P}_C(\cdot) = \mathbb{P}(\cdot|C)$ defines a probability measure. Comparing (2.14) with (2.11), we see that conditional independence given C is nothing but ordinary independence with respect to $\mathbb{P}_C$. In symbols, A and B are conditionally independent given C if and only if $A \perp\!\!\!\perp_{\mathbb{P}_C} B$.

Example 2.21.

- (i) Unconditional independence is the same as conditional independence given Ω (or given any C with $\mathbb{P}(C) = 1$).
- (ii) Consider the events A , B and C in Example 2.19. Note that $A \cap B \cap C = \emptyset$, so

$$0 = \mathbb{P}(A \cap B|C) \neq \mathbb{P}(A|C)\mathbb{P}(B|C) = \frac{1}{2} \cdot \frac{1}{2}.$$

Thus A and B are conditionally dependent given C even though they are unconditionally independent.

- (iii) Consider in the context of Table 1.1 the following events:

$$\begin{aligned} A &= \{+++, ++-, +-+, +--\} = \{\text{first toss is } +\}, \\ B &= \{++-, +-+, -++\} = \{\text{two } +\}, \\ C &= \{+-+, +- -, -++, -+-\} = \{\text{one } + \text{ and one } - \text{ in first two tosses}\}. \end{aligned}$$

Note that

$$\mathbb{P}(A \cap B) = \frac{2}{8} \neq \frac{4}{8} \cdot \frac{3}{8} = \mathbb{P}(A)\mathbb{P}(B).$$

So A and B are (unconditionally) dependent.

But A and B are conditionally independent given C :

$$\mathbb{P}(A \cap B|C) = \frac{1}{4} = \frac{2}{4} \cdot \frac{2}{4} = \mathbb{P}(A|C)\mathbb{P}(B|C).$$

Let's explain intuitively why this is the case. If C occurs, there is exactly one $+$ in the first two tosses. Suppose also B occurs. Then the last toss must be $+$. Now the first two tosses are either $+-$ or $-+$. Both cases are equally likely (given C). So A and B are conditionally independent given C .⁵

2.5 Random variable

So far we have the following intuitive picture: in an experiment, a sample point $\omega \in \Omega$ is picked according to the underlying probability measure $\mathbb{P}$. This picture is not complete. In many applications, the sample point ω itself may not be directly observed. Also, we may be interested in some specific quantities depending on the outcome. We illustrate with some examples:

- Suppose we play a game of cards. Let the sample point capture the complete state of the deck after shuffling. Several cards are then dealt (and observed). Now we must decide what to do with only partial information about (the rest of) the deck. Note that the game may end without seeing all cards.
- Suppose we perform a *randomized survey*.⁶ The quantity of interest may be the proportion of respondents who support a given policy.
- Suppose we model the price of a stock over the next 10 days. The stock price S_t at each time t is a random quantity.

In probability theory, we use *random variables* to model quantities of interest that arise in the experiment. Depending on the context, those quantities may take values in different *state spaces*.

Definition 2.22 (Random variable). *Given a state space $\mathcal{X}$, an $\mathcal{X}$ -valued random variable is a function X from the sample space Ω to its state space $\mathcal{X}$, i.e., $X : \Omega \rightarrow \mathcal{X}$. Given an outcome $\omega \in \Omega$, we call $X(\omega)$ the realized value of X .*

Typically, the state space $\mathcal{X}$ is a subset of $\mathbb{R}$ or $\mathbb{R}^d$. When $\mathcal{X} \subset \mathbb{R}$, we often call X a *random variable* without the qualifier “real-valued” (this is very common in practice). If $\mathcal{X} \subset \mathbb{R}^d$, we call X a (d -dimensional) *random vector* and write $X = (X_1, \dots, X_d)$. Then each X_i is a real-valued random variable, i.e., $X_i : \Omega \rightarrow \mathbb{R}$. It is also common to call $X = (X_1, \dots, X_d)$ a random vector even if some of the components are not real-valued. Since different

⁵Hidden in this example is a fundamental concept in probability theory, namely the *Markov property*. We might touch upon it towards the end of the course.

⁶This means that we interview a number of randomly chosen individuals. This itself involves many design choices which are beyond the scope of this course.

conventions are used by different authors, you should be careful about the state spaces of the random variables under discussion. Sometimes we use the boldface $\mathbf{X}$, as in $\mathbf{X} = (X_1, \dots, X_d)$, to stress that $\mathbf{X}$ is a random vector.

Remark 2.23. The state space $\mathcal{X}$ of X may be strictly larger than the *range* $X(\Omega) = \{X(\omega) : \omega \in \Omega\}$. For example, if the only possible values of X are 0 and 1 we can still call X a real-valued random variable.

Example 2.24 (Some examples of random variables). Suppose we flip a coin n times and let the sample space be $\Omega = \{0, 1\}^n$. Each sample point $\omega = (\omega_1, \dots, \omega_n)$ is a binary vector with length n : $\omega_i \in \{0, 1\}$. Note that coin tossing is simply a convenient language. The sample space Ω can be used to model any chance event in which we observe n binary outcomes. For example, if we observe the price of a stock over n days, we may suppose $\omega_i = 1$ means that the stock price increases on day i (and $\omega_i = 0$ otherwise).

Here are some random variables defined on Ω :

- (i) $X : \Omega \rightarrow \{0, 1, \dots, n\}$ defined by $X(\omega) = \omega_1 + \dots + \omega_n$ is the total number of ones. Then $\frac{1}{n}X$ is the proportion (i.e., the relative frequency) of ones. It is a natural estimator of, say, the probability of heads.
- (ii) $Y : \Omega \rightarrow \{0, 1, \dots, n\}$ defined by $Y(\omega) = n - X(\omega)$ is the total number of zeros.
- (iii) $Z : \Omega \rightarrow \{1, \dots, n\}$ defined by $Z(\omega) = \max\{k \in \{1, \dots, n\} : \omega_i = \dots = \omega_{i+k-1} = 1 \text{ for some } i\}$ if $\sum_i \omega_i > 0$ and $Z(\omega) = 0$ if $\sum_i \omega_i = 0$ is the length of the longest consecutive ones.
- (iv) Suppose for simplicity that $n = 2k$ is even. Define $U = (U_1, U_2) : \Omega \rightarrow \mathbb{R}^2$ by $U_1(\omega) = \frac{1}{k}(\omega_1 + \dots + \omega_k)$ and $U_2(\omega) = \frac{1}{k}(\omega_{k+1} + \dots + \omega_{2k})$. Then U is a bivariate random vector which gives the proportions of ones in the first half and the second half. It is a natural random variable to consider if one suspects that the data generating process is “*nonstationary*”.

Given a random variable $X : \Omega \rightarrow \mathcal{X}$, it is natural to consider events defined in terms of X . For example, in Example 2.24(i) we may be interested in the event that $X = 2$. More generally, given a subset $B \subset \mathcal{X}$, we may consider the event

$$\{X \in B\} := \{\omega \in \Omega : X(\omega) \in B\}.$$

Mathematically, $\{X \in B\} = X^{-1}(B)$ is the *preimage* of the set B . The probabilities of such events define the *distribution* of X .

Example 2.25. In the context of Table 1.1, suppose $B = \{0, 1\}$. Then

$$\begin{aligned} \{X \in B\} &= \{X \in \{0, 1\}\} = \{\omega \in \Omega : X(\omega) \in \{0, 1\}\} \\ &= \{- - -, - - +, - + -, + - -\}. \end{aligned}$$

Note that letting $B = (-\infty, 1]$ results in the same event (as a subset of Ω). When a random variable is real-valued, events of the form $\{X \leq x\}$, where x ranges over $\mathbb{R}$, are very useful as will be seen later.

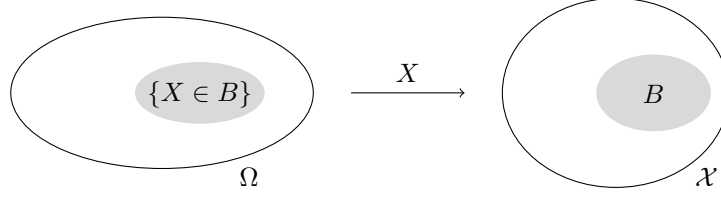


Figure 2.1: A random variable X is a function from the sample space Ω to its state space $\mathcal{X}$. For $B \subset \mathcal{X}$, the event $\{X \in B\}$ is the preimage $X^{-1}(B)$ which is a subset of Ω .

Definition 2.26 (Distribution). *On a given probability space $(\Omega, \mathbb{P})$, let $X : \Omega \rightarrow \mathcal{X}$ be an $\mathcal{X}$ -valued random variable. The distribution of X is the set function $\mathbb{Q}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}\{\omega : X(\omega) \in B\}$ defined for $B \subset \mathcal{X}$.*

Given an experiment described by the probability space $(\Omega, \mathbb{P})$ and a random variable $X : \Omega \rightarrow \mathcal{X}$, “observing the value of X ” can be regarded as a new experiment whose sample space is the state space $\mathcal{X}$ of X . As shown by the following theorem, the distribution of X is the probability measure on $\mathcal{X}$. So $(\mathcal{X}, \mathbb{Q}_X)$ is a new probability space induced by $(\Omega, \mathbb{P})$ and $X : \Omega \rightarrow \mathcal{X}$. The proof gives us another chance to practice with Definition 2.7.

Theorem 2.27. *The distribution $\mathbb{Q}_X$ of a random variable $X : \Omega \rightarrow \mathcal{X}$ is a probability measure on $\mathcal{X}$.*

Proof. We verify that $\mathbb{Q}_X$ satisfies the properties in Definition 2.7:

- (i) (normalization) We have $\mathbb{Q}_X(\mathcal{X}) = \mathbb{P}(X \in \mathcal{X}) = \mathbb{P}(\Omega) = 1$.
- (ii) (sum rule) Let $B_1, B_2, \dots$ be a sequence of disjoint sets in $\mathcal{X}$. For each i , let $A_i = \{X \in B_i\} = X^{-1}(B_i)$ which is an event in Ω . Since $B_1, B_2, \dots$ are disjoint, so are $A_1, A_2, \dots$. It follows that

$$\begin{aligned} \mathbb{Q}_X\left(\bigcup_{i \geq 1} B_i\right) &= \mathbb{P}\left(X \in \bigcup_{i \geq 1} B_i\right) = \mathbb{P}\left(\bigcup_{i \geq 1} \{X \in B_i\}\right) \\ &= \mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i) = \sum_{i \geq 1} \mathbb{Q}_X(B_i). \end{aligned}$$

□

We say that a random variable $X : \Omega \rightarrow \mathcal{X}$ is *discrete* if there exists a discrete subset $S = \{x_1, x_2, \dots\}$ (or $S = \{x_0, x_1, \dots\}$ depending on the indexing convention) of $\mathcal{X}$ such that $\mathbb{P}(X \in S) = 1$. Intuitively, this means that the set of strictly possible values of X is discrete. In this case, the distribution of X is completely characterized by the individual probabilities $p_i = \mathbb{P}(X = x_i)$ since for any $B \subset \mathcal{X}$ we have

$$\mathbb{Q}_X(B) = \mathbb{P}(X \in B) = \sum_{i: x_i \in B} p_i. \quad (2.15)$$

The probability $\mathbb{P}(X = x)$, as a function of x , is called the *probability mass function* of X .

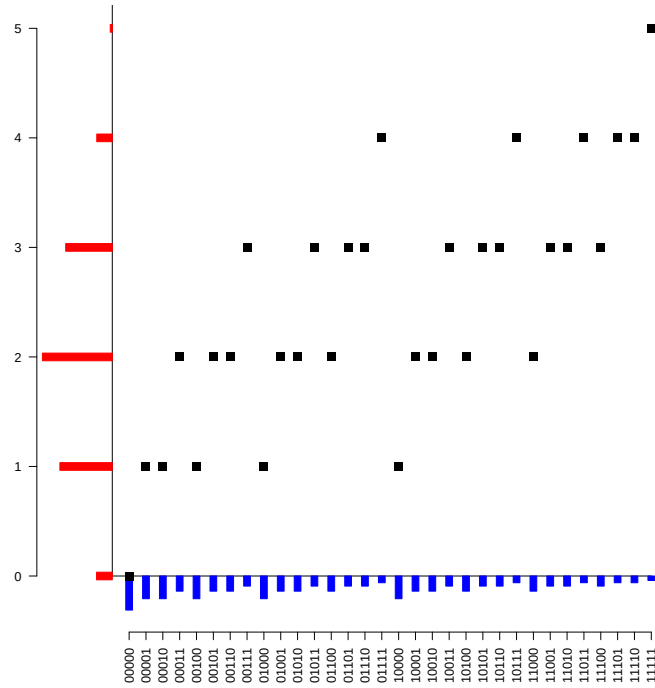


Figure 2.2: Tossing a biased coin 5 times (Example 2.30) where the probability of heads (indicated by 1) is 0.4. The vertical blue bars give the probabilities $\mathbb{P}\{\omega\}$ for all $\omega \in \Omega$. Let X be the total number of heads. The graph of X is indicated by the black dots. The horizontal red bars give the probability mass function of X .

Definition 2.28 (Probability mass function). *Let X be a discrete random variable. The function $f_X : \mathcal{X} \rightarrow [0, 1]$ defined by $f_X(x) = \mathbb{P}(X = x)$ is called the probability mass function (pmf) of X .*

From (2.15), if we know the pmf f_X we know how to compute $\mathbb{Q}_X(B)$ for any B . Thus we say that f_X characterizes $\mathbb{Q}_X$. For random variables which are not discrete we need other tools, such as *probability density functions*, to describe their distributions. Probability density functions, which characterize the distributions of *continuous* random variables, will be introduced in Section 4. We end this section with some examples of probability mass functions.

Example 2.29. Consider once again Table 1.1. Let X be the number of + as in the table. Then X is a discrete random variable since $\mathbb{P}(X \in \{0, 1, 2, 3\}) = 1$. The probability mass function of X is given by

$$f_X(x) = \begin{cases} 1/8, & \text{if } x = 0; \\ 3/8, & \text{if } x = 1; \\ 3/8, & \text{if } x = 2; \\ 1/8, & \text{if } x = 3; \\ 0, & \text{otherwise.} \end{cases}$$

Example 2.30. In Figure 2.2 we consider tossing a coin 5 times. The sample space is $\Omega = \{0, 1\}^5$ and its elements are enumerated on the x -axis. For convenience we write $\omega_1\omega_2\omega_3\omega_4\omega_5$ instead of $(\omega_1, \omega_2, \omega_3, \omega_4, \omega_5)$. Here $\mathbb{P}$ is defined by the requirement that the probability of getting 1 (heads) for each toss is 0.4 (so the coin is biased) and that the different tosses are *jointly independent*. For example, $\mathbb{P}\{01001\} = 0.4 \cdot 0.6 \cdot 0.4 \cdot 0.4 \cdot 0.6 = 0.02304$ is the probability of observing the sequence 01001 (in that specific order). The precise definition of joint independence will be given in Section 4. Here we simply note that it is essentially an extension of the product rule (2.11) to multiple events. Let X be the total number of heads. In the figure we show (i) $\mathbb{P}\{\omega\}$ for each ω , (ii) the graph of $X : \Omega \rightarrow \{0, 1, \dots, 5\}$ and (iii) the probability mass function of X .

The distributions in both Example 2.29 and 2.30 are special cases of the very important *binomial distribution* (see Definition 4.26 below).